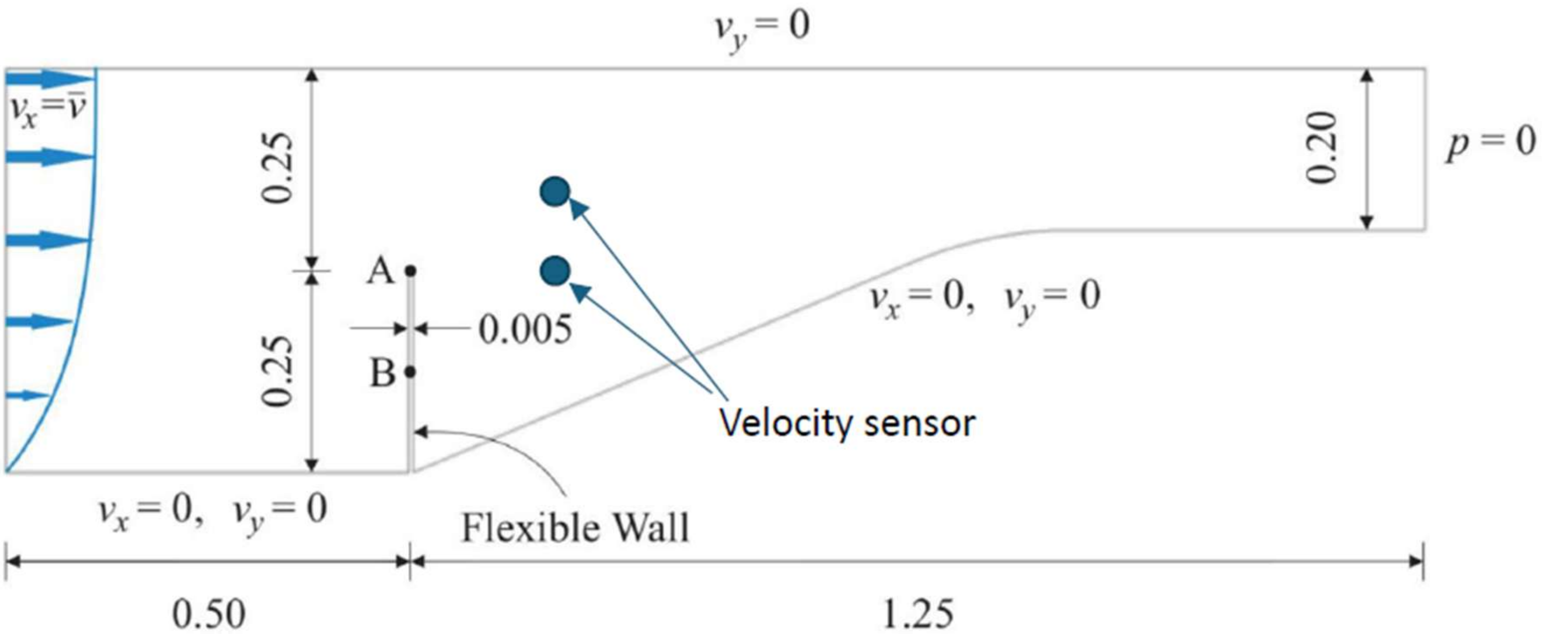
Technische Universität Braunschweig

Developing Methodology for Model Fitting to be Used in Fluid-Structure Coupled Problem

Bhakti Sheth
Rohin Ramisetty
Shrinidhi Holenarsipura

Supervisor: Ihar Antonau

Problem Statement



Tasks:

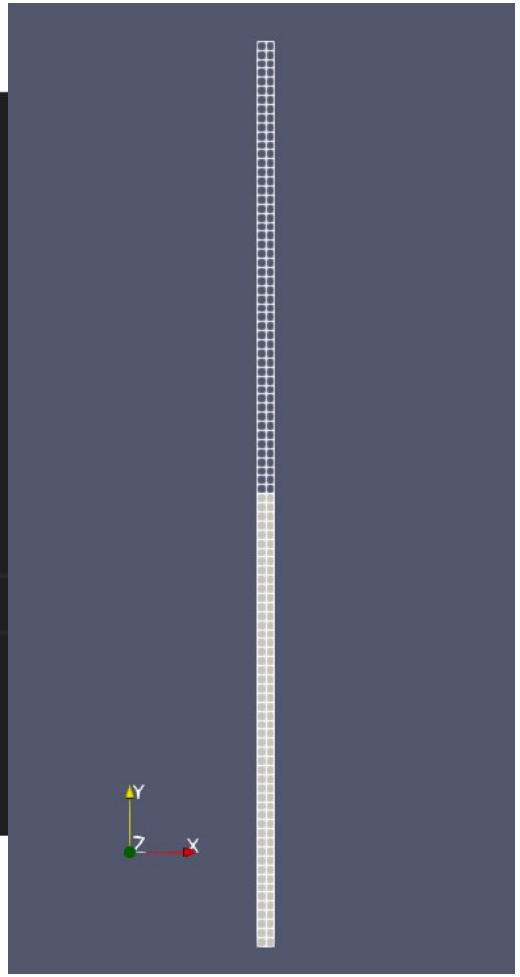
1. Understand the coupled problem.
2. Get the interested values from the model at the given locations.
3. Compute the displacement / velocities (i.e. measured in x direction) of the two sensors with prescribed Young's moduli for E_1, E_2, E_3.
4. Build a response function to compute the sensor error between statistical quantities (i.e. average) of measured and computed from the simulation (i.e. $h_i^m(t), h_i^c(t)$).

$$J = \frac{1}{2}\sum_{i=1}^{M}\sum_{t=1}^{N}[h_i^c(t) - h_i^m(t)]^2$$

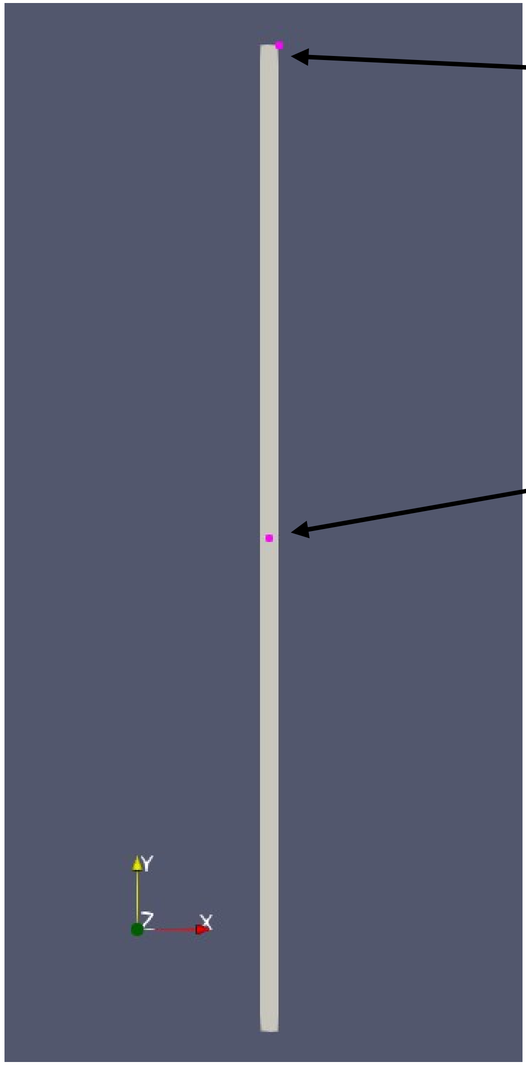
5. Developing an optimization process to identify correct thermal parameters.
6. (Studying the effect of different coupling methods on the system identification.)

Mesh Manipulation

```
//    SubModelPart: top_half
//      Number of Nodes: 154
//      Number of Elements: 100
//      Number of Conditions: 0
//      Number of Properties: 2
//      Number of SubModelParts: 0
//    SubModelPart: bottom_half
//      Number of Nodes: 150
//      Number of Elements: 100
//      Number of Conditions: 0
//      Number of Properties: 2
//      Number of SubModelParts: 0
```



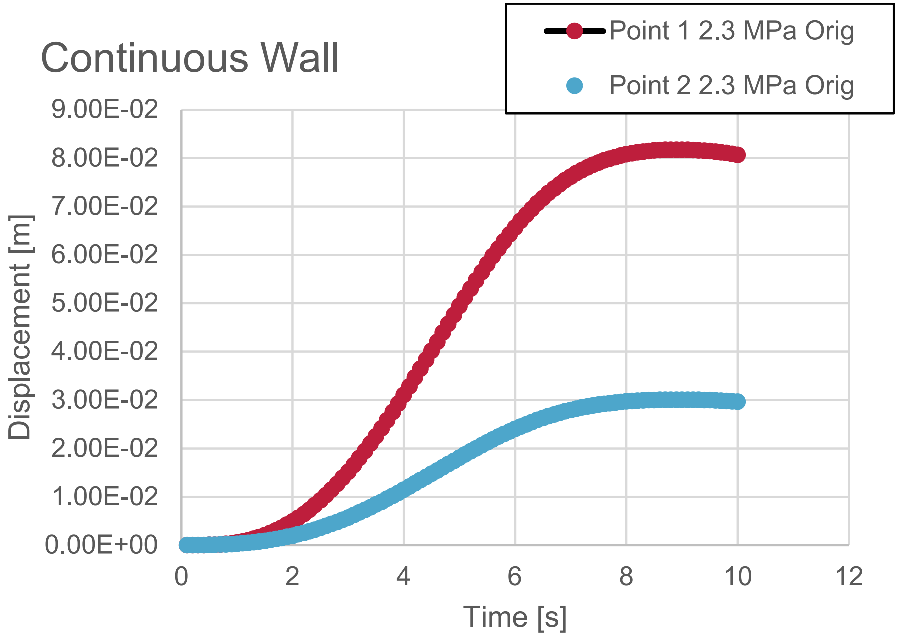
Displacement Sensors

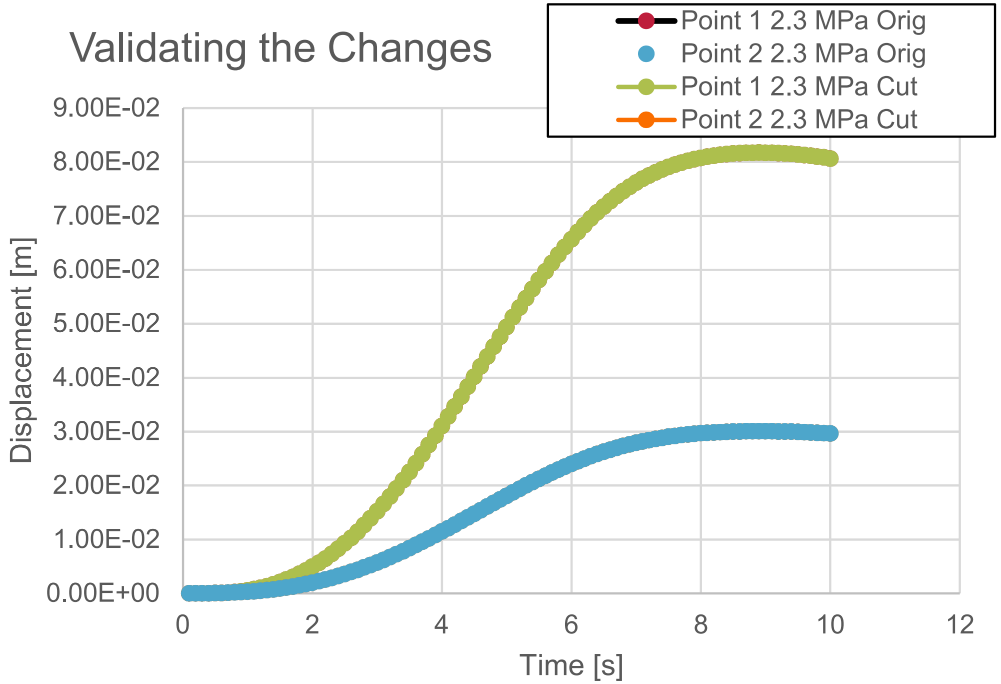


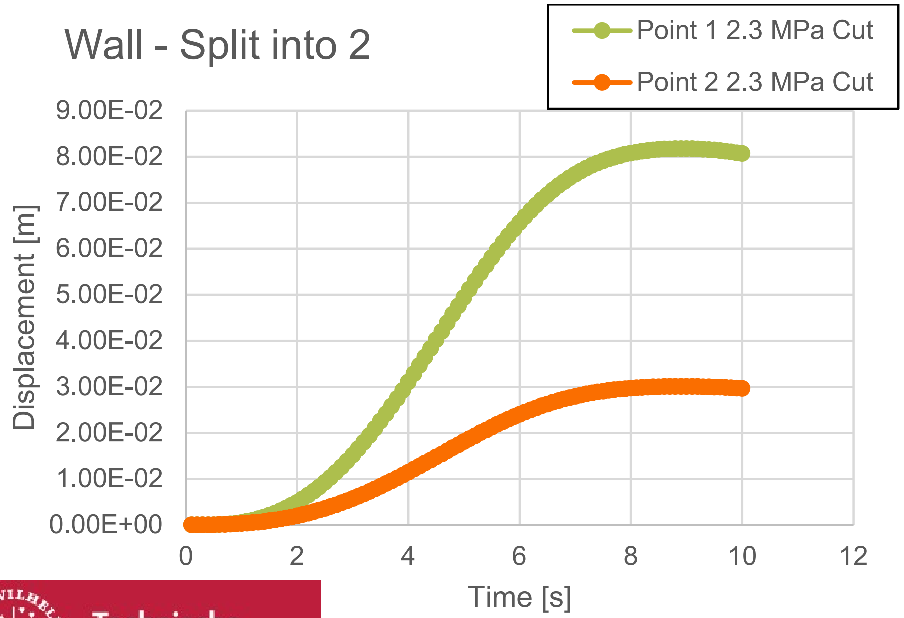
Point 1 [0.5015,0.25,0.0]

Point 2 [0.5005, 0.1261, 0.0]

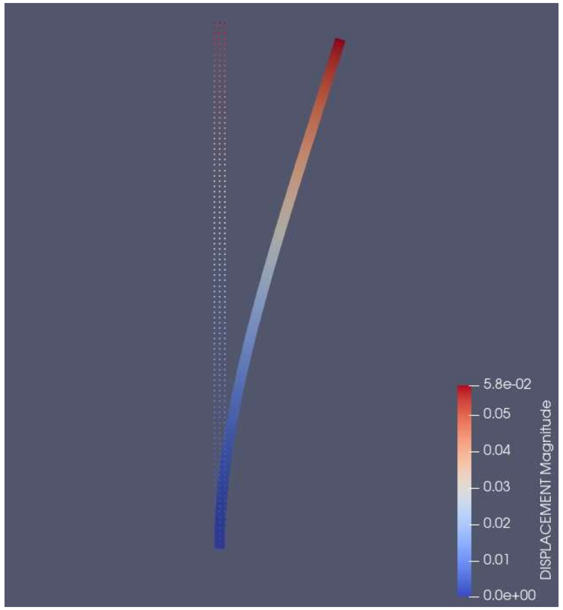
Computation and Comparision - Displacement







Computation and Comparision - Displacement



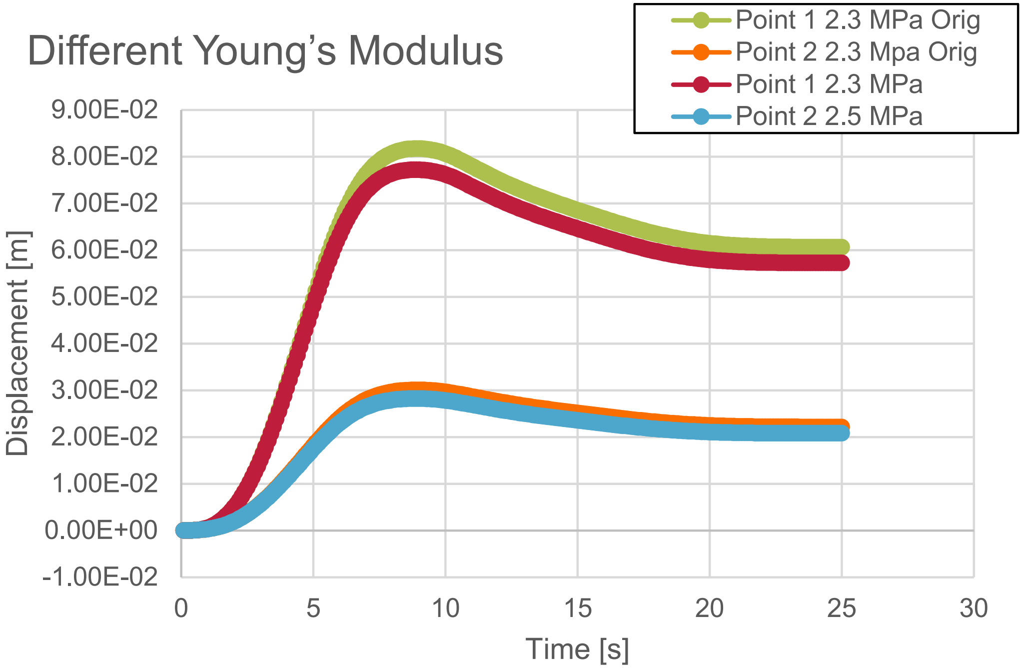


Fig: Bottom is Stiffer than the Top

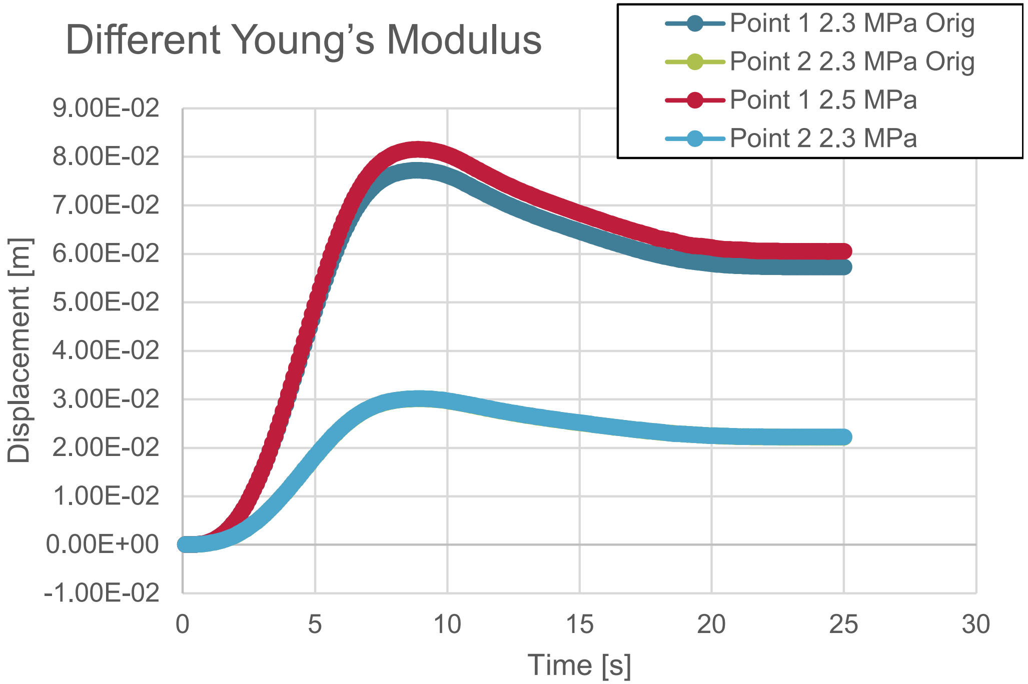
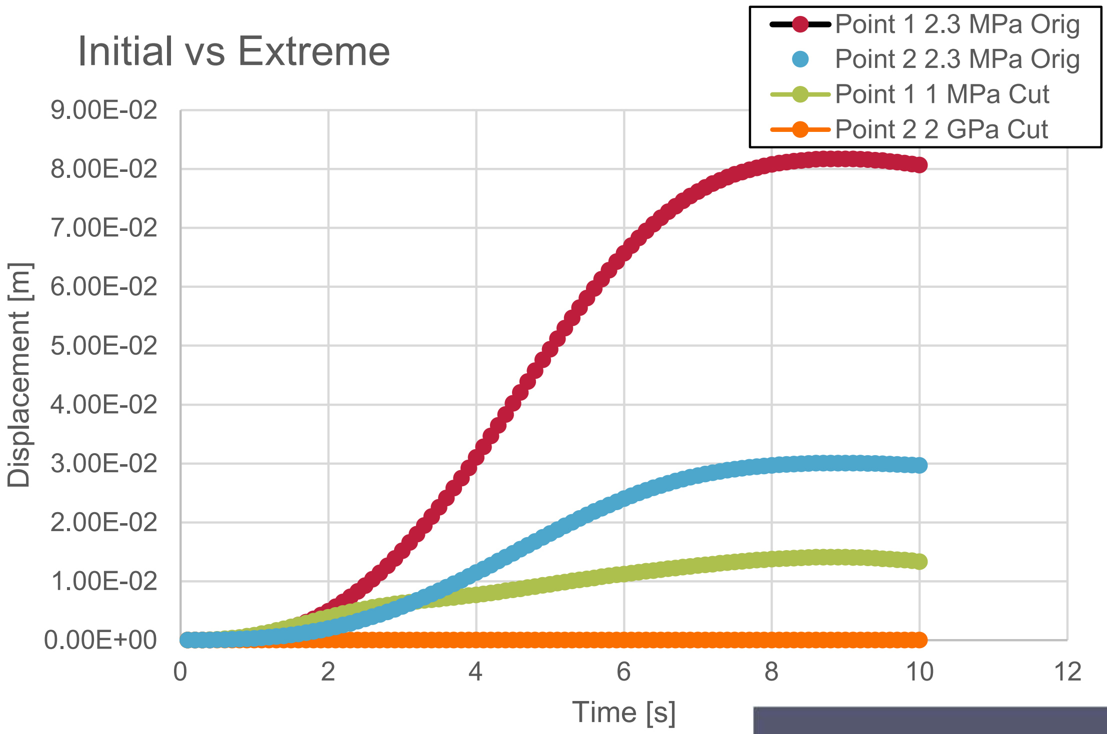


Fig: Top is Stiffer than the Bottom

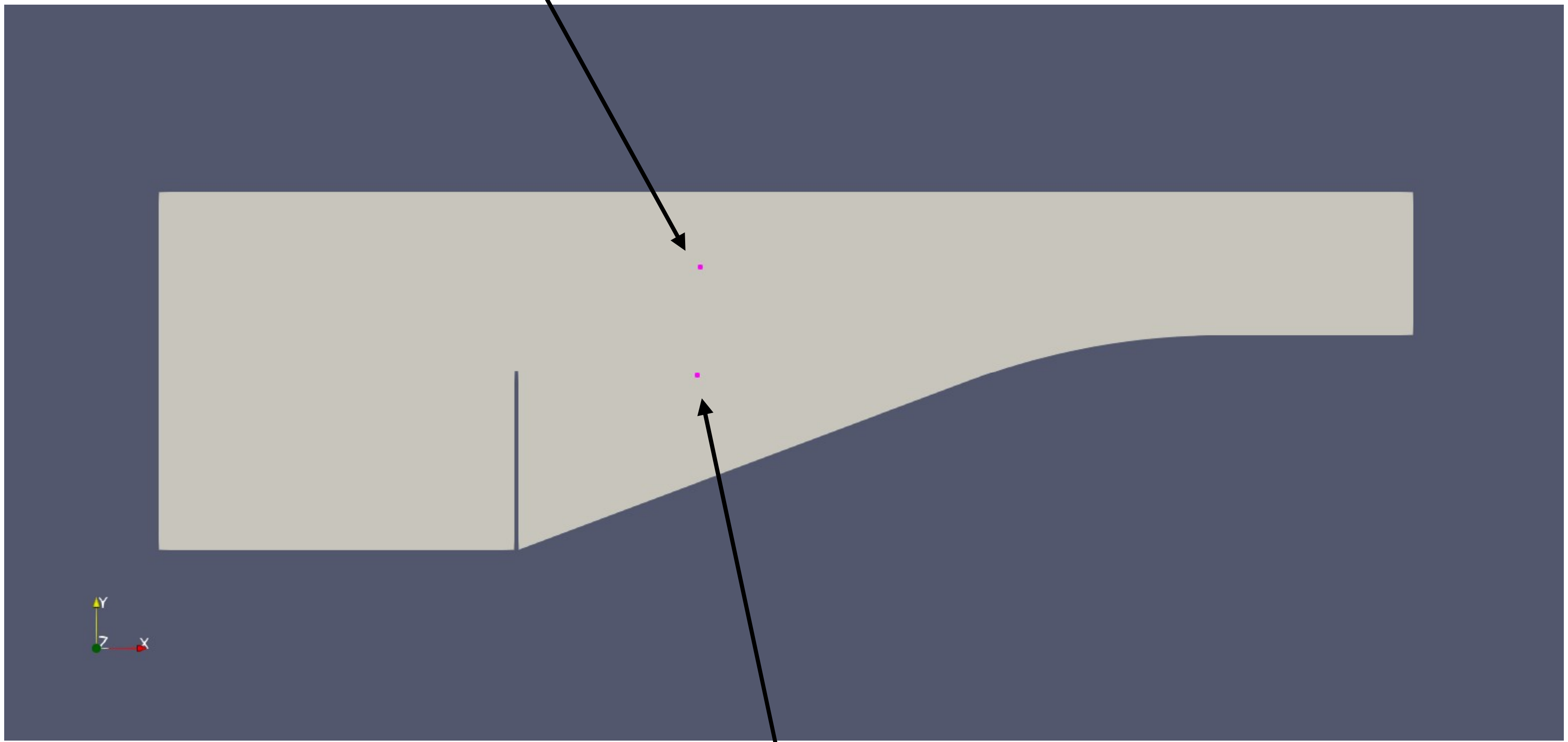
Computation and Comparision - Displacement



Technische Universität Braunschweig

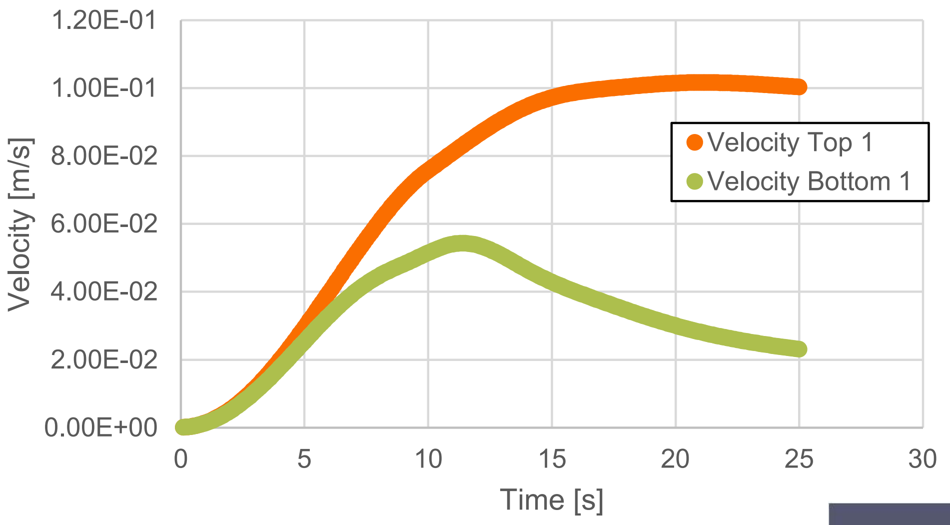
Velocity Sensors

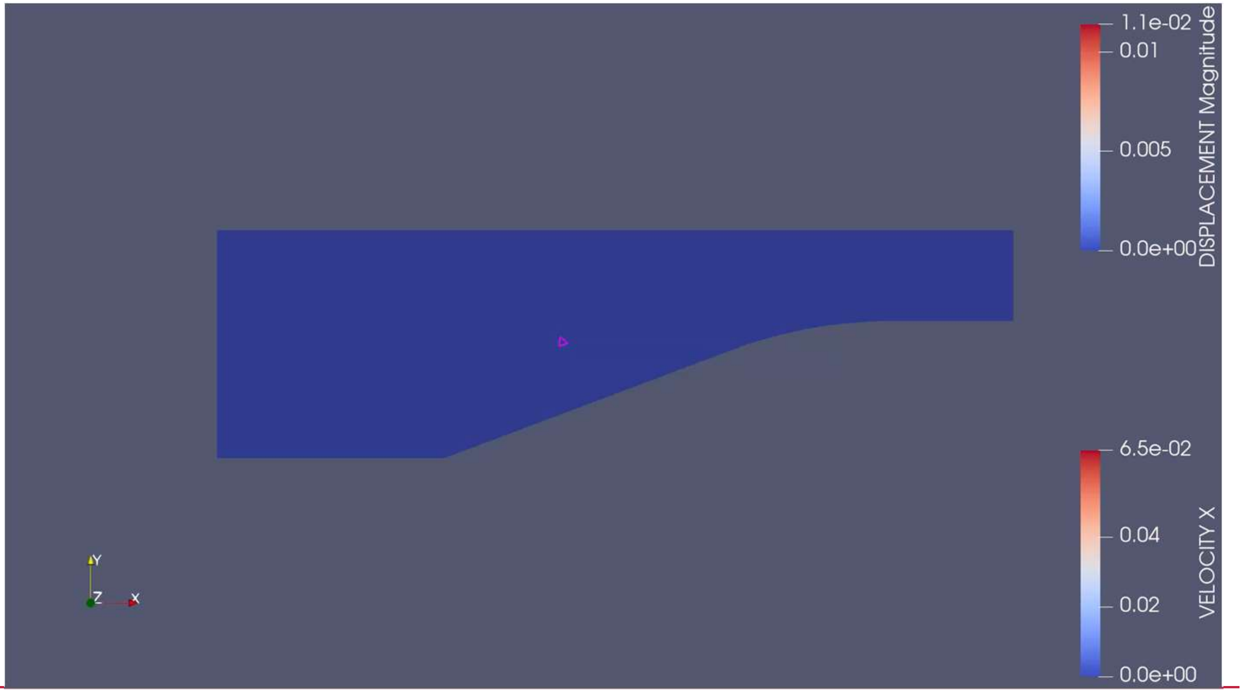
[0.75, 0.25, 0.0]



[0.75, 0.40, 0.0]

Computation - Velocity



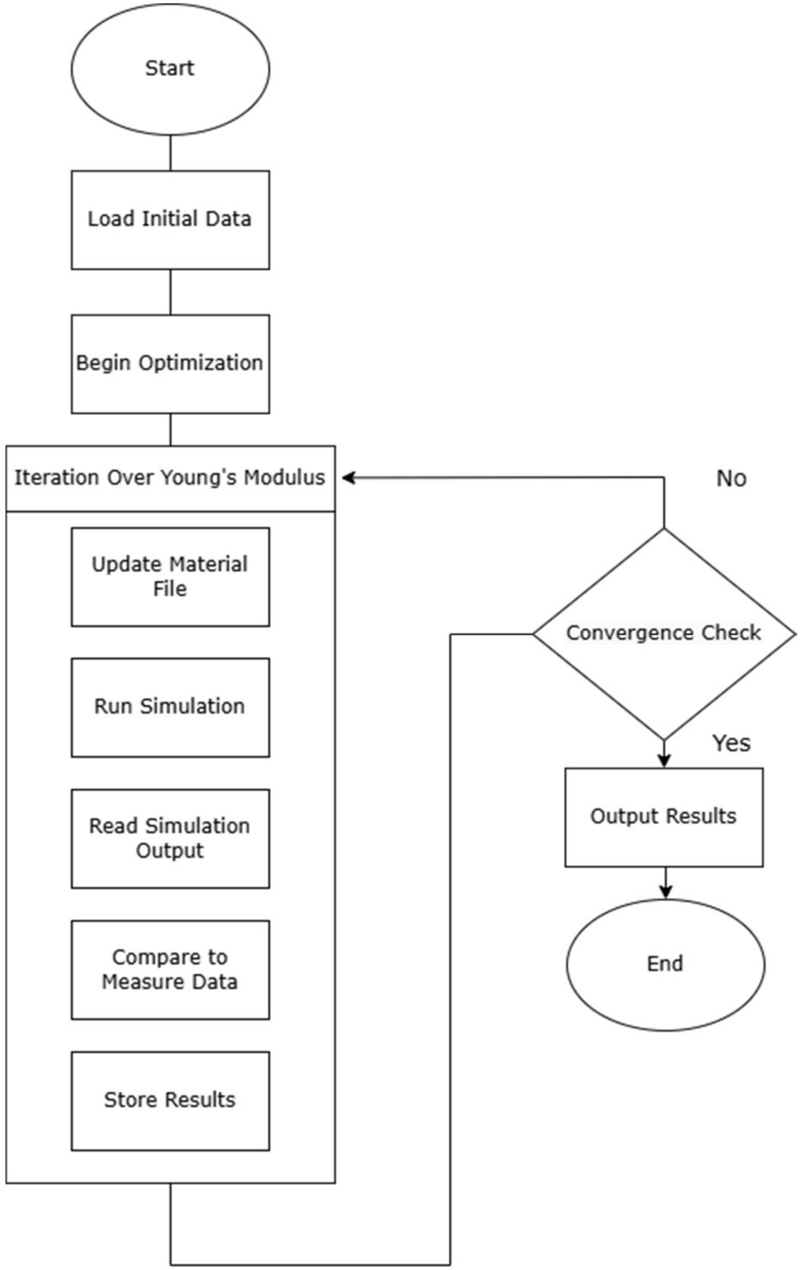


Optimization Process

$$J = \frac{1}{2}\sum_{i=1}^{M}\sum_{t=1}^{N}[h_i^c(t) - h_i^m(t)]^2$$

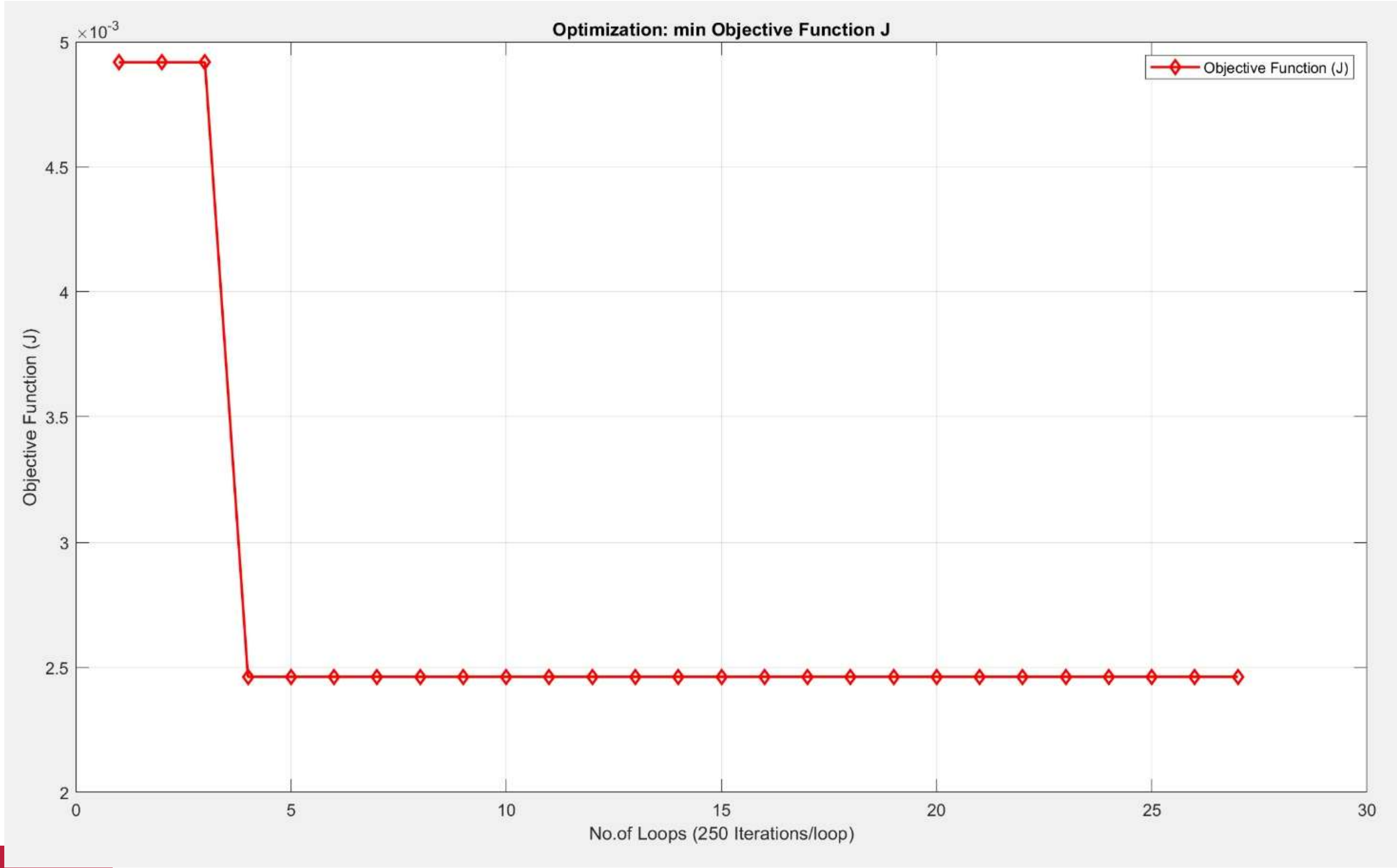
J: Objective Value
M: number of sensors
t: number of timesteps
hc = average computed values of displacement/veloctiy sensor

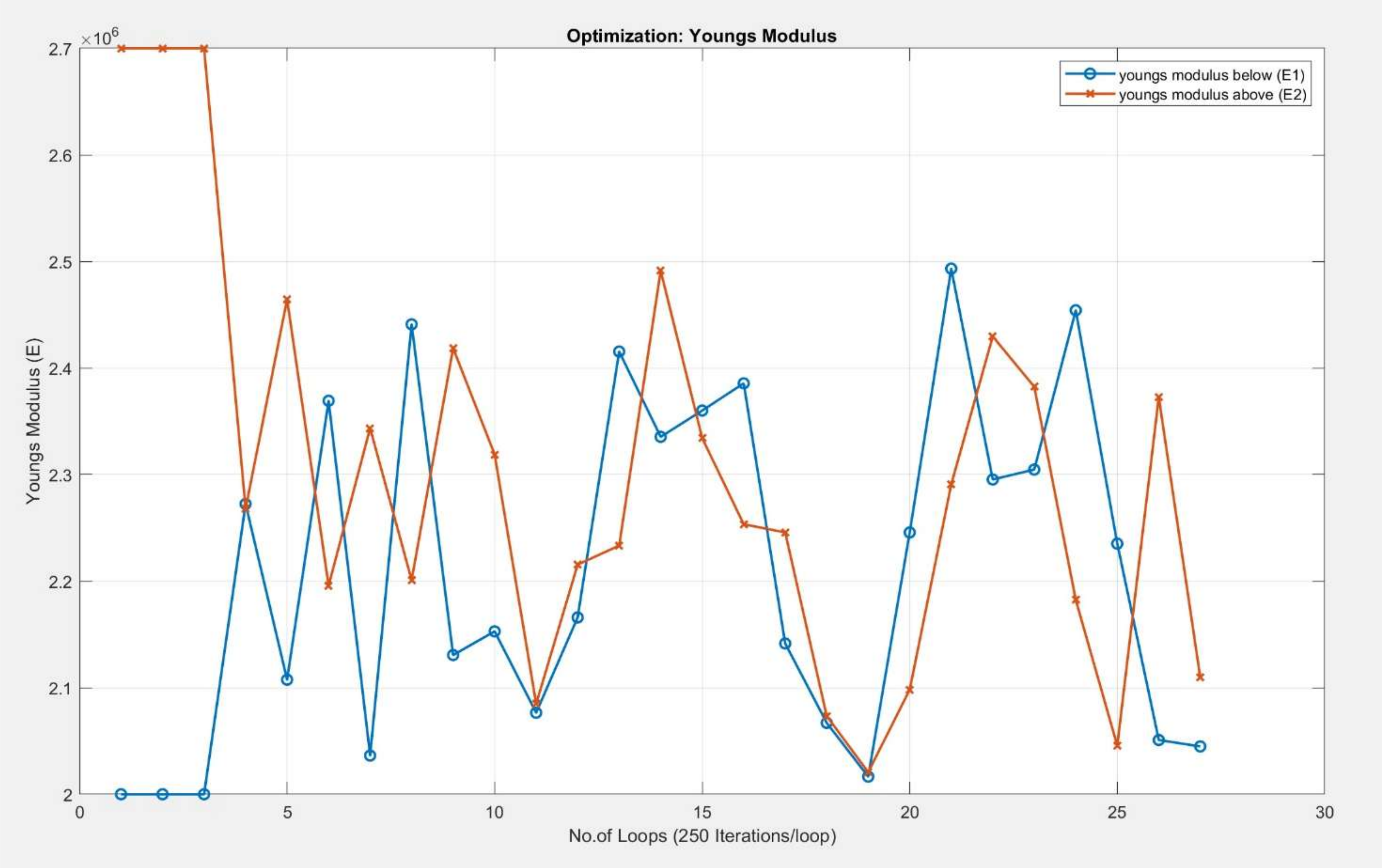
Hm = average measured values of displacement/veloctiy sensor



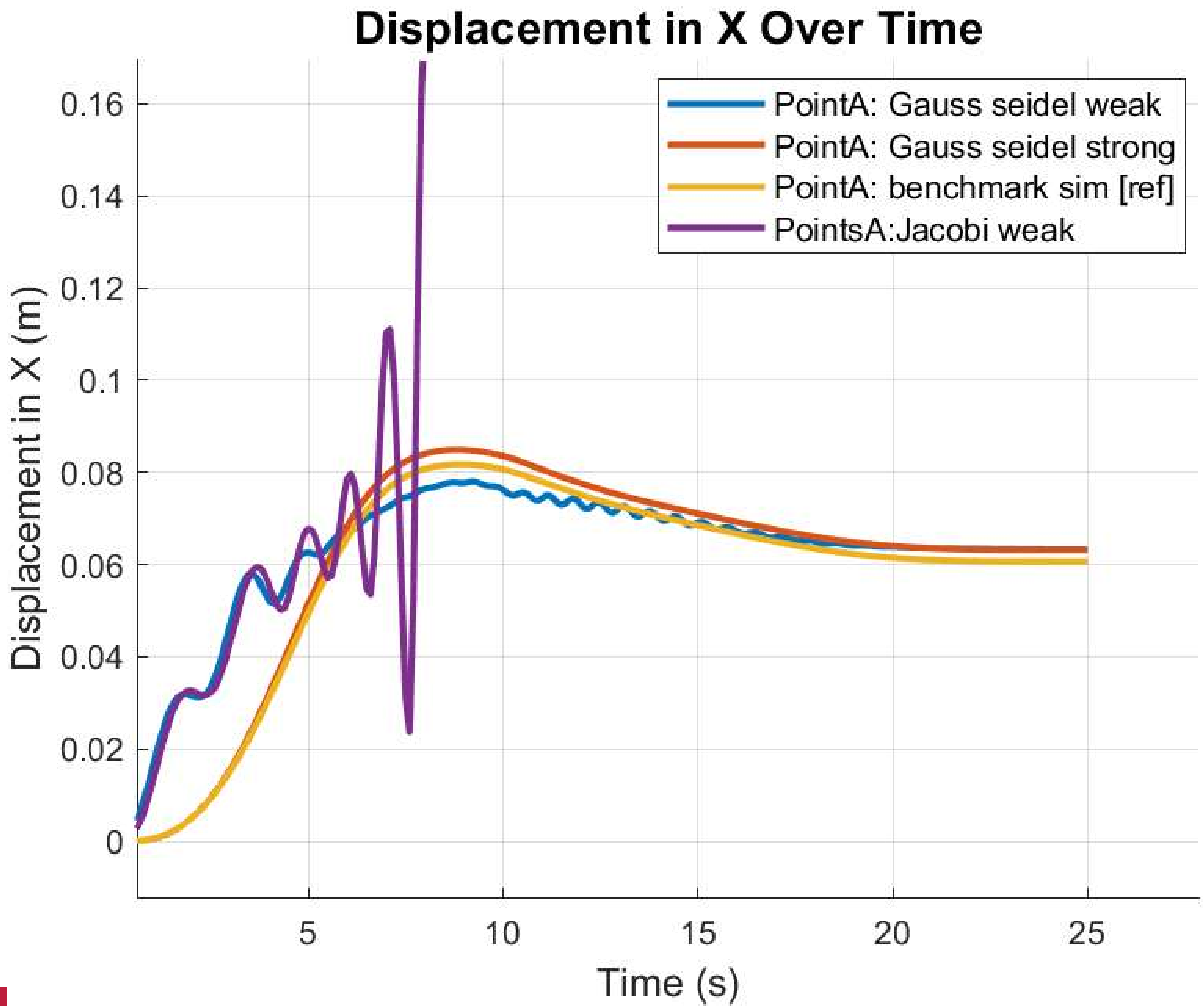
Type of optimizers

scipy.optimize.**differential_evolution**
Coupled solver: gauss seidel weak with 4 sensors (displacement/velocity)




Optimization: Youngs Modulus
×10^6
2.7
2.6
2.5
2.4
2.3
2.2
2.1
2
Youngs Modulus (E)
0
5
10
15
20
25
30
No.of Loops (250 Iterations/loop)
youngs modulus below (E1)
youngs modulus above (E2)

Comparitive Study – Coupling Methods



Thank you for listening.

Splitter Function

Logic used for the split function

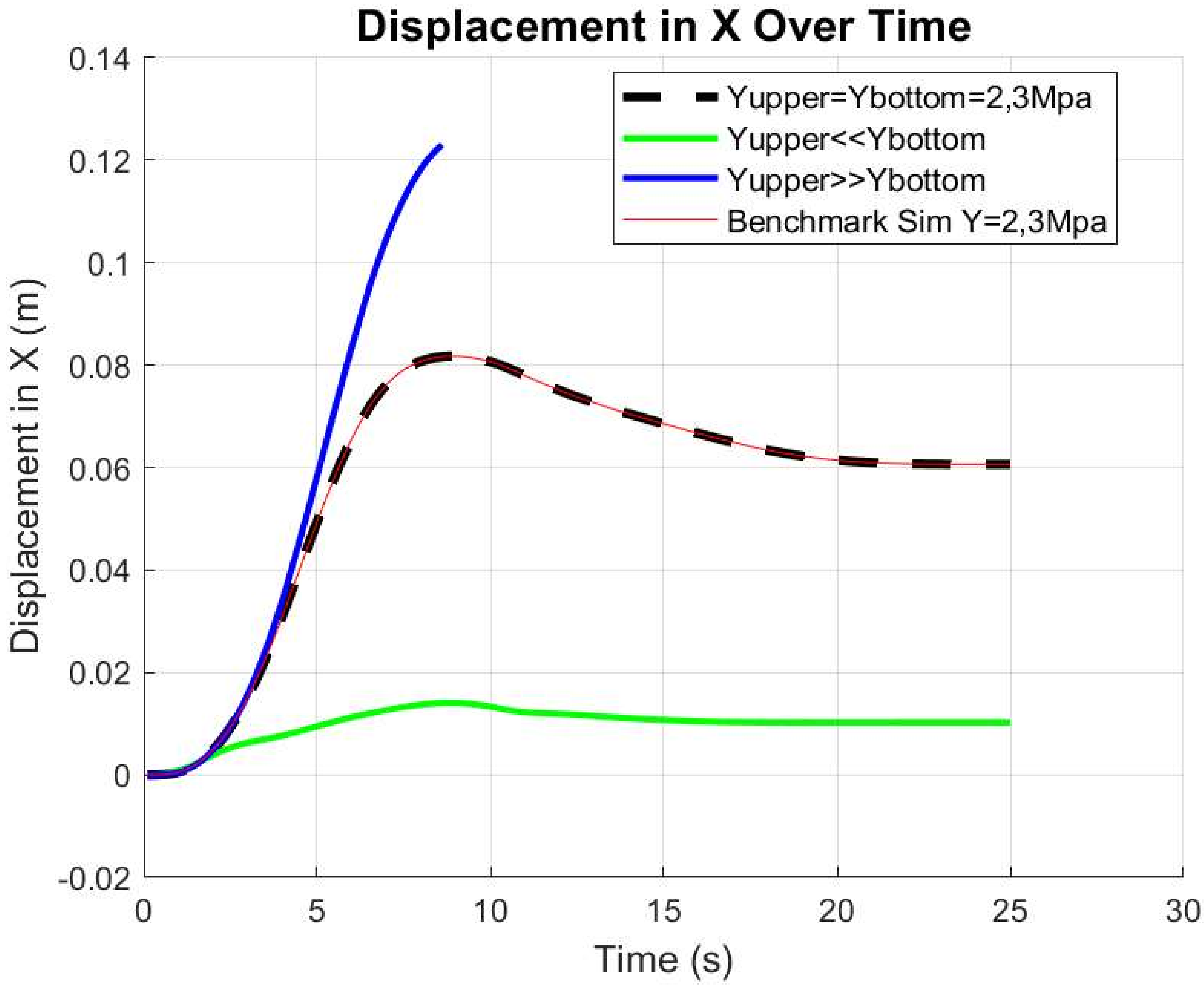
Interface nodes:

- Input required parameters: divisions d
  - *If d=1: exit*
- Read CSM.mdpa
  - *N={n,x,y,z}*
  - *Find y min*
  - *Find y max*
- Calculate H= ymax-ymin
  lenght of sub element Ls = H/d
- No.of split points Ns = d-1
- Set y_split=0
- initialise matrix for interface I=[Ns][3]
- loop for i to Ns, increment in i
  - y_split=y_split+Ls
  - {loop for j from 0 to 3, increment in j}
    - If y_split==N(y)

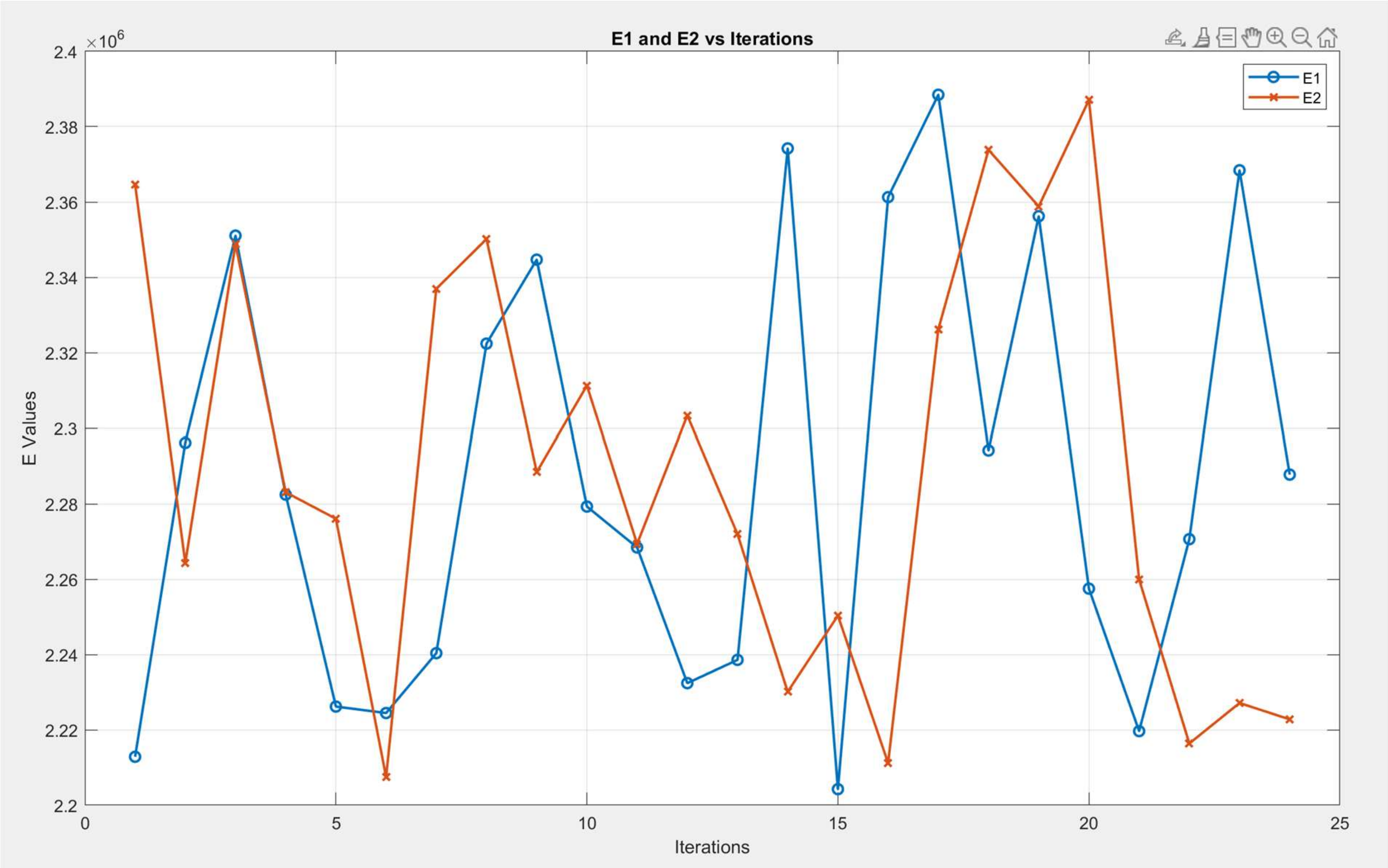
nodes in each submodel.

- set y submodel=0
- loop k from 1 to Ns, increments in K
  - y_submodel= y_submodel + Ls
  - For y in range of [y_submodel, y @ N{k}]
    - Submodel{k}= N{n}

- *Can divide beam into n equal parts*
- *Can also divide beam in unequal parts, [User needs to specify a ratio]*

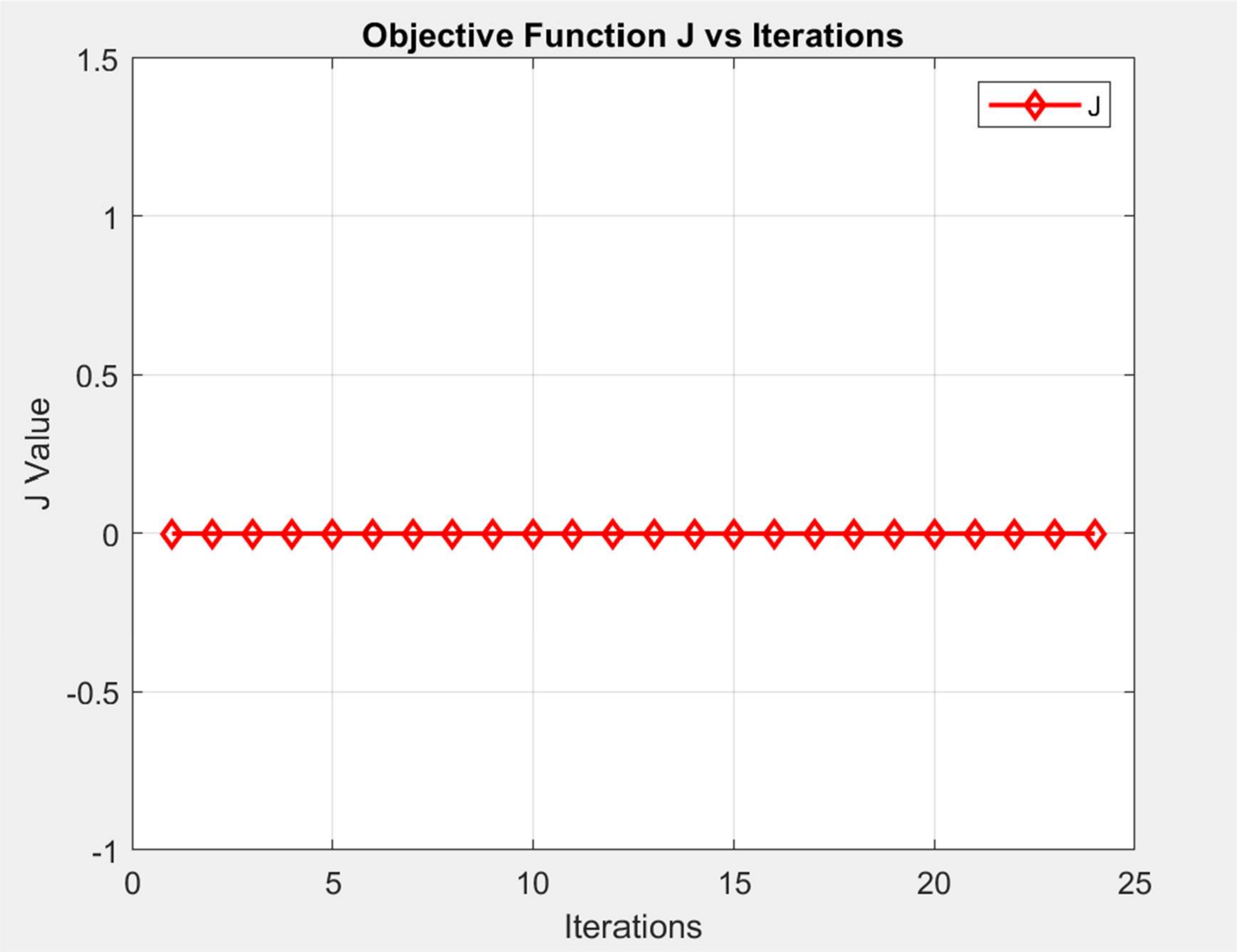

Displacement in X Over Time
Yupper=Ybottom=2,3Mpa
Yupper<<Ybottom
Yupper>>Ybottom
Benchmark Sim Y=2,3Mpa
Displacement in X (m)
Time (s)
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
-0.02
0
5
10
15
20
25
30

Optimization with only displacement sensors



Technische Universität Braunschweig

Optimization with only displacement sensors



Gauss Seidel vs Jacobi solving strategies visualised